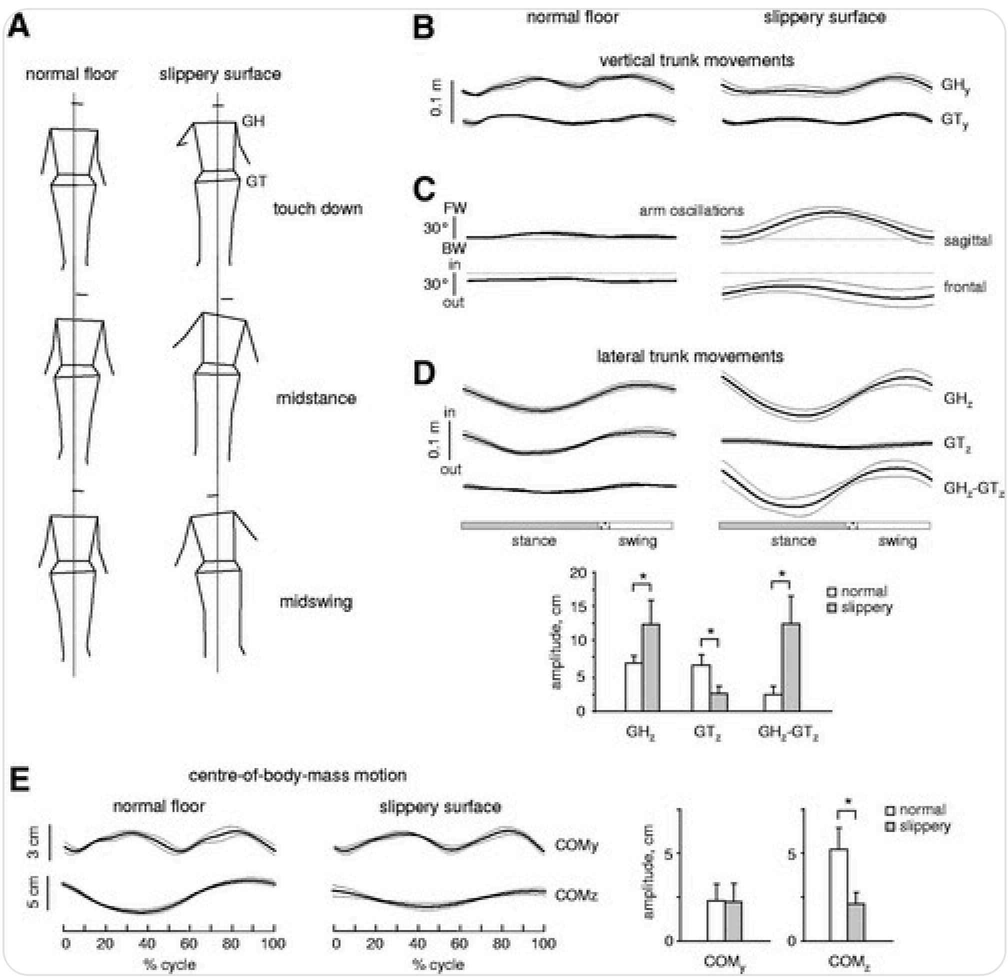
Copy of Penguin Plan for Summer 2026

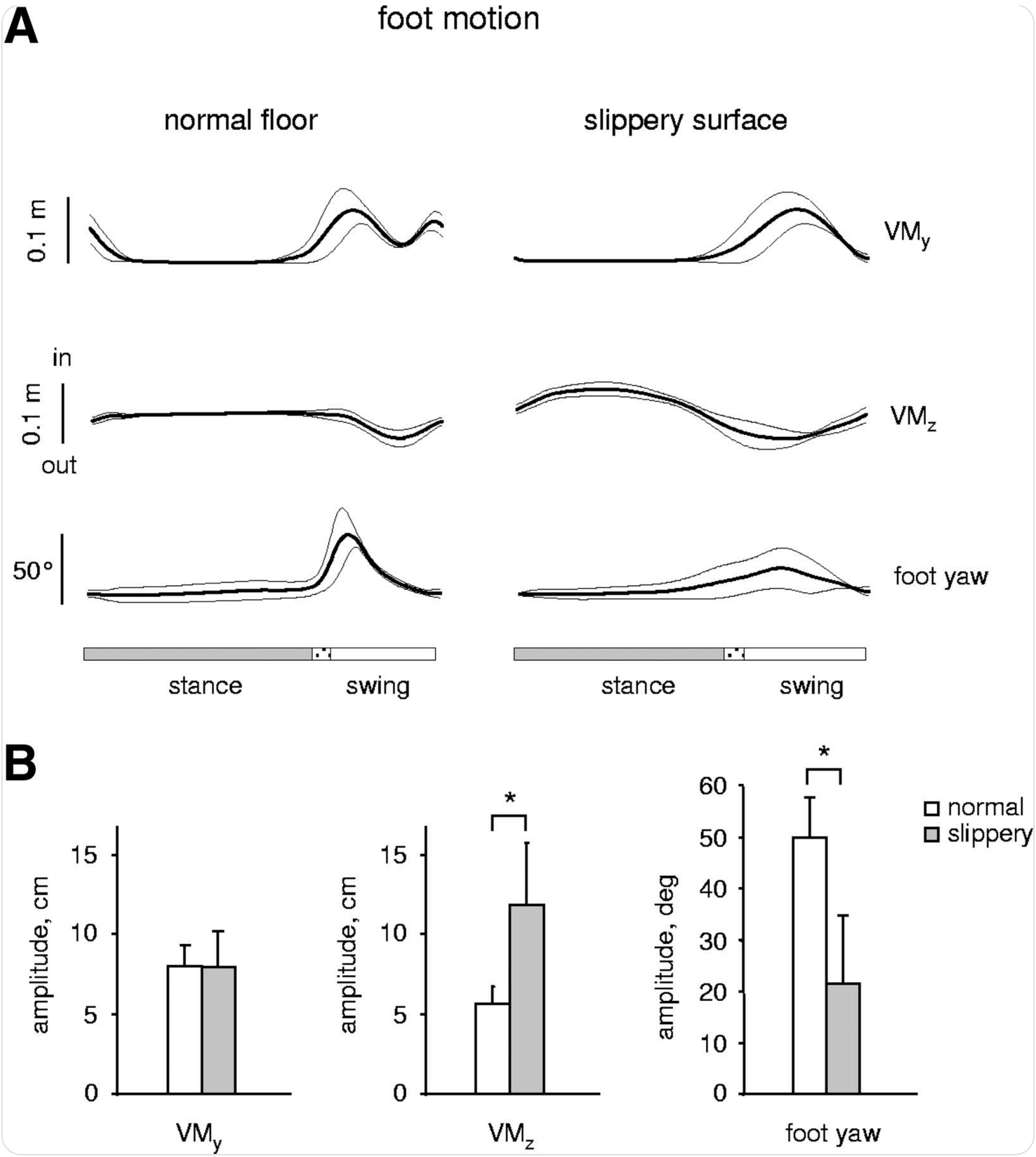
- **Claim for the paper:** Penguin-like walking where the torso is over the stance foot is better for slippery surfaces (based on the direction of the GRF). Penguins with their torsos tilted over the stance foot and humans walk the same way ONLY slippery surfaces based on "Motor Patterns During Walking on a Slippery Walkway" - https://journals.physiology.org/doi/full/10.1152/jn.00499.2009?rfr_dat=cr_pub++0pubmed&url_ver=Z39.88-2003&rfr_id=ori%3Arid%3Acrossref.org. The paper refers to this as nonplantigrade walking i.e. the humands aren't walking with the regularly "flat human foot rolling on the ground", but like a bird where there is more lateral foot (roll) motion and less forward foot (pitching) motion. Humans on non-slippery surfaces, don't show torso roll (they walk with no torso phasing/tilting i.e. the torso is held upright).



- We will test the robot with different torso phasing and mass distribution to understand bipedal walking on slippery surfaces and see how these two parameters can inform walking. For the human mass distribution, we want the COM to be between 54-57% of the full height when standing ("Determining the location of the body's center of mass for different groups of physically active peopl" -

https://www.sciencedirect.com/science/article/pii/S0021929014002103?via%3Dihub)

- **Sim experiments** will be combinations of penguin/human mass distributions with penguin/non-plantigrade slippery surface/no torso phasing. (For Ben to do)
    - These are the sim experiments:
        - Torso held upright (PD for no swinging) with penguin mass distribution
        - Torso held upright (PD for no swinging) with human mass distribution
        - Just lower body (no torso)
        - Penguin-like torso swinging (torso swinging over stance leg) with penguin mass distribution
        - Penguin-like torso swinging (torso swinging over stance leg) with human mass distribution
        - Nonplantigrade-like non-slippery surface torso swinging (torso swinging over swing leg) with penguin mass distribution
        - Nonplantigrade-like non-slippery surface torso swinging (torso swinging over swing leg) with human mass distribution
    - Each sim will be optimized (you can just do a param sweep) to get the best amplitude and frequencies for the motors. We will look at GRFs and look at friction cone (normal and tangential forces measured while walking) to determine the minimum friction required for walking. Additionally, we will report out walking speed and COM. We will also look at foot roll vs foot pitch on slippery vs non slippery surfaces (we should check if there is a single peak in foot motion).



- **The hardware experiments** will be run with the full human configuration (human mass distribution and torso swinging), the full penguin configuration (penguin mass distribution and torso swinging), torso held upright for both penguin and human experiments, and no torso. The controls will remain the same for either penguin or human mass distribution. Naomi will measure walking and COM on multiple surfaces.